

CSCE 717 ALGORITHMIC GAME THEORY · MODULE 06 · LECTURE NOTES

Mechanism Design

Choosing the rules so that honesty is optimal.

MODULE OUTCOMES

- State the mechanism design problem.
- Define incentive compatibility and individual rationality.
- Derive the VCG mechanism and explain its payments.
- State the revelation principle and what it simplifies.
- State the impossibility results that bound the field.

1 The problem

CHOOSE AN ALLOCATION RULE AND A PAYMENT RULE SO THAT REPORTING TRUTHFULLY IS OPTIMAL FOR EVERY AGENT

Agents have private types — valuations, costs, preferences, capacities — **and report something to the mechanism; the mechanism maps the reports to an outcome and to a payment for each agent.** Those two rules are the whole design.

Incentive compatibility: reporting the true type is optimal for each agent — in dominant strategies if possible, which requires nothing at all about what the other agents do (Module 02 §2), and in Bayes-Nash equilibrium if not, which requires a lot more.

Individual rationality: participating must be at least as good as not participating — otherwise agents simply leave and the mechanism is irrelevant however elegant its properties are.

And then the designer's objective: efficiency (maximise total value), revenue, or some fairness criterion — and the payments are the instrument that buys the incentive property, which is why mechanisms without money (Modules 09 and 11) are so much more constrained.

2 The revelation principle

ANYTHING IMPLEMENTABLE BY ANY MECHANISM IS IMPLEMENTABLE BY A TRUTHFUL DIRECT ONE

Given any mechanism and the agents' equilibrium strategies in it, construct a new mechanism that asks each agent for its type and then plays that agent's equilibrium strategy on its behalf — the outcome is identical, and now reporting honestly is optimal, because the simulation does the strategising.

So the search for mechanisms can be restricted to truthful direct mechanisms without any loss of generality — which is an enormous simplification and is a large part of why the field is tractable at all: the space of all possible rule systems is unmanageable, and this cuts it to a space with structure.

And it means that an impossibility proved for truthful mechanisms holds for all mechanisms — which is exactly what gives §4's results their force, and is why they are stated about truthful mechanisms without that being a restriction.

But note the practical caveat: the constructed mechanism requires the designer to know the agents' equilibrium strategies in order to simulate them — and **in practice indirect mechanisms, such as ascending auctions, are considerably easier for real people to participate in** than a direct revelation of a full valuation function (Module 08 §4).

3 VCG

$\text{payment}_i = (\text{best total value for the others without } i) - (\text{the others' value in the chosen allocation with } i)$

- **Allocate so as to maximise the total reported value** — the efficient allocation, computed as though the reports were true.
- **And charge each agent the externality it imposes on everybody else:** the difference between how well the others would have done without this agent and how well they do in the chosen allocation — **which is Module 05 §2's marginal cost fix, arriving as a fully general mechanism.**
- **So agent i 's net utility equals the total value with i minus the total value without i — and the second term does not depend on i 's report at all**, so i maximises its utility exactly by making the mechanism maximise the first term, **which truthful reporting does. That is the entire proof**, and it is worth reconstructing once.
- **VCG is the externality idea, generalised** — and recognising it as the same idea as marginal cost pricing makes both easier to remember and to apply.
- **Truthful in dominant strategies, and efficient** — **which is a great deal to obtain simultaneously**, and is why VCG is the field's central construction and the benchmark everything else is compared against.
- **But it requires computing the optimal allocation — $n + 1$ times, once overall and once with each agent removed — which is NP-hard in combinatorial settings** (Module 08 §1) and is therefore frequently not available.
- **And approximating the allocation breaks truthfulness** — **the proof above uses exact optimality in an essential way** — **which is the central tension of Module 08** and one of the most productive problems in the field.
- **Plus it is vulnerable to collusion and to false-name bidding** (one agent bidding under several identities), **and its revenue can be zero or very low** in plausible cases, which matters enormously to anybody actually running an auction.
- **Which is why VCG is used much less often than its theoretical standing would suggest — a useful corrective**, and a good instance of a theoretically dominant solution losing to practical considerations (Module 13 §2).

4 The limits

MYERSON-SATTERTHWAITE: NO MECHANISM FOR BILATERAL TRADE IS EFFICIENT, TRUTHFUL, INDIVIDUALLY RATIONAL, AND BUDGET BALANCED

One buyer and one seller, each with a private value for the good — you cannot have all four properties at once, and the proof is constructive about why: the payments required to make both sides truthful do not balance, so somebody must subsidise the trade.

So every real trading mechanism gives one of them up — an outside subsidy (a platform fee structure that runs a deficit on some trades), some inefficiency (trades that should happen and do not), or some manipulability — and knowing which one is exactly how to evaluate a real mechanism.

And Gibbard-Satterthwaite is the ordinal analogue (Module 10 §3): without money, truthfulness over a sufficiently rich preference domain forces dictatorship — which is why money matters so much in this field, and is not a moral claim but a structural one.

Which is the field's shape in one sentence: with money, a great deal is possible and VCG is the general answer; without money, very little is — and Modules 09 and 11 work within that constraint rather than escaping it.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Nisan et al., chapters 9 and 11 (free PDF)	§§1 to 3 — mechanism design and VCG, with the computational constraints treated seriously.
Vickrey, Clarke, and Groves — the three original papers	§3 — and it is worth seeing how differently the three authors arrived at the same construction.
Myerson & Satterthwaite — Efficient mechanisms for bilateral trading	§4 — the impossibility, in the original.
Roughgarden — Twenty Lectures, lectures 2 to 7 (free)	The whole module — and the development of VCG from single-item auctions is the clearest route to it.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **State the four components** of a mechanism design problem.
- 2 **Give an example** where individual rationality binds.
- 3 **Prove the revelation principle** in your own words.

- 4 **Explain why it strengthens impossibility results.**
- 5 **Compute VCG payments** for a three-agent, two-item example.
- 6 **Verify truthfulness** by checking one agent's misreports.
- 7 **Construct a VCG instance with zero revenue.**
- 8 **Show that false-name bidding profits** in some VCG instance.
- 9 **State Myerson-Satterthwaite** and identify which property three real marketplaces give up.
- 10 **Explain why money matters**, using the two impossibility results.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 State the mechanism design problem's four components.
- 2 Define incentive compatibility, and say which form is preferable.
- 3 Why does individual rationality matter?
- 4 State the revelation principle and what it buys.
- 5 Why does it strengthen impossibilities?
- 6 Give VCG's allocation and payment rules.
- 7 Sketch the truthfulness proof.
- 8 Name four costs of VCG.
- 9 State Myerson-Satterthwaite and its four properties.
- 10 What is the field's shape, with and without money?